

Research Article

Multi-modal vision-based deformable perception for in-finger manipulation with soft active surfaces

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ABSTRACT

Enabling robots to manipulate objects dexterously within their grasp often requires complex multi-fingered hands with dense, multi-modal sensor arrays. This complexity can limit practicality and robustness, especially in confined spaces. This study challenges that paradigm by introducing a robust multi-modal perception method that derives 6D forces and poses from the large-scale deformation of a soft finger, using only a single internal camera. We propose ActiveSPN2, a robot gripper with a soft active surface designed for in-finger manipulation in constrained environments, featuring multi-modal perception through a vision-based deformable perception architecture. By tracking the real-time deformation of the internal Soft Polyhedral Network, we achieve real-time, multi-modal vision-based perception that provides 6D forces and object pose estimation during in-finger manipulation. Quantitatively, the proposed force perception model achieves an average accuracy of approximately 0.286 N on the test set, indicating reliable generalization under unseen manipulation conditions. A series of experiments were conducted on the ActiveSPN2 gripper using the proposed vision-based deformable perception approach, demonstrating the effectiveness and robustness of the strategy.

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1. Introduction

This study investigates the feasibility of performing force and pose estimation to achieve multi-modal perception on an active surface, as the robotic hand's dexterity and tactile perception are two significant factors for achieving efficient and robust in-finger manipulation. Some common tasks, such as screwing a light bulb into a socket [1,2], require the robot to move the bulb in a way that allows the robot to exert the torque required to screw the bulb in. In addition to screwing in a light bulb, picking up an object from a shelf [3–5] also changes the position and posture of the object and maintains the grasping force. Combined, many real-world manipulation tasks require human-like dexterity and perception capability, where the robot may be required to exert a force through a grasped object or may have geometric constraints on how the object is positioned. This capability is particularly valuable for in-hand manipulation with a small workspace because moving objects by the robotic manipulator requires a large workspace [6,7]. The applications of in-finger manipulation in confined spaces are common in a wide range of everyday activities, including shelf picking, box sorting, and pipeline maintenance.

The multi-articulated dexterous hand [8] is a typical design for manipulators that perform in-hand manipulation, demonstrating high dexterity. With tactile films providing sparsely distributed force information [9], they can efficiently learn and perform dexterous policies for specific skill sets. To capture more detailed tactile information from the contacted object, the widely used vision-based tactile sensor GelSight [10], capable of reconstructing high-resolution 3D contact geometry, has been integrated into dexterous hands [11]. However, equipping the fingers and palm with full tactile perception remains challenging, given the difficulty of scaling vision-based tactile sensors to a compact form [12].

Compared with multi-fingered dexterous hands, roller grippers with continuous active surfaces [13–16] demonstrate comparable or even superior dexterity and efficiency in some tasks. BACH [13] showcases an active surface design with high stability and dexterity, but due to the complexity of its continuous structure, it currently lacks sensing capabilities. Leveraging the advanced dexterity of roller grippers and the optical principles of GelSight for tactile perception, the Tactile-Reactive Roller Gripper [14] enables efficient continuous rolling of unknown objects. Building on the rotational fingertip design and the high-resolution vision-based tactile perception, TacRot [15] enables precise in-hand reorientation of objects, allowing pivoting and twisting motions while maintaining continuous tactile feedback. By integrating rotatable high-resolution tactile sensors, RoTip-Bot [16] enables grasping, feeding, and counting of multiple

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Table 1

Comparison of active belt/roller-driven mechanism in sensing capabilities and manipulation performance.

	Sensing capability	Sensing method	Geometric adaptation	In-hand Manip
BACH [13]	×	×	2D-adaptive	✓
Roller Grasper V3 [14]	×	×	×	✓
TacRot [15]	Localization	Photometric Stereo Tech	At Fingertip	2 Dof
RoTipBot [16]	Force	Photometric Stereo Tech	At Fingertip	1 Dof
TouchRoller [17]	Localization	Photometric Stereo Tech	Surface	1 Dof
ActiveSPN [21]	Pose Estimation	Internal Vision	Omni-adaptive	✓
ActiveSPN2 (Ours)	VBDormP: Force +Pose	Internal Vision	Omni-adaptive	✓

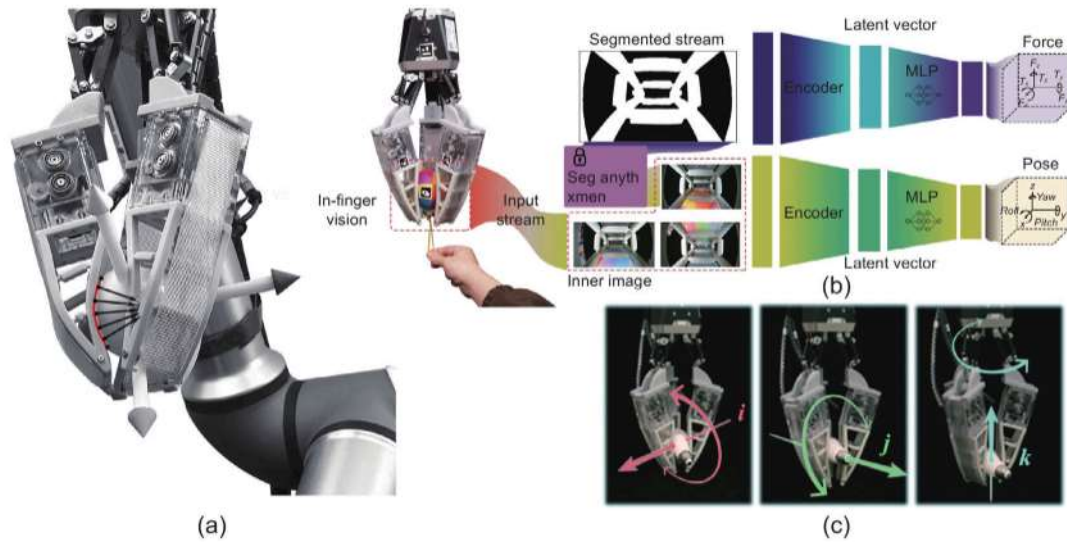


Fig. 1. Multi-modal vision-based deformable perception of soft active surface with an omni-adaptive surface for in-finger manipulation. (a) The ActiveSPN2 finger with an active surface installed on existing grippers. (b) Overview of the force and pose estimation learning architecture. (c) In-finger manipulation of grasped objects via ActiveSPN2.

layers of thin, flexible objects while maintaining continuous tactile perception throughout the process. However, these grippers, which employ the photometric stereo technique [10] used in GelSight, require uniform, colorful illumination from multiple directions parallel to the surface. Meeting this requirement is extremely challenging for a continuous active surface moving along the surface direction. Moreover, the cylindrical roller shape reduces the tactile contact area, diminishing stability during object manipulation and limiting dexterity.

Besides dexterous hands, previous studies of in-hand manipulation using vision-based tactile sensors can be categorized into three main types: passive rollers [17,18], actively driven rollers [14–16,19], and continuous active surfaces [20,21]. A comparative summary of these representative designs in terms of sensing capabilities and manipulation performance is provided in Table 1. In addition to the commonly used transparent elastomers that employ the photometric stereo technique, the use of semitransparent elastomers has also been explored [20,22]. The method using a semitransparent elastomer employs a translucent coating that permits partial external light transmission, allowing for the perception of the surrounding scene prior to contact. However, the acquired images are typically blurred due to light scattering within the coating.

In current research, soft active surfaces have primarily focused on vision-based tactile perception. In contrast, for soft robots without active surfaces, researchers have attempted to embed a large number of sensors within the soft material, including resistive [23], capacitive [24], and piezoelectric tactile sensors [25], stretchable strain sensors [26], optical fibers [27], or conductive composites [28,29], to capture detailed deformation information of the robotic body [30]. Such multi-modal perception aims to address the challenges introduced by continuous

and high-dimensional deformations in robotic modeling and control. Although this approach can enhance observability to some extent, these embedded sensors often compromise the mechanical properties of the soft material, significantly increase system complexity [31], and are difficult to integrate into continuous rotation mechanisms due to wiring constraints and the risk of tangling on rotating surfaces. Therefore, there is a strong need for stable and robust perception methods for soft active surfaces that enable dexterous manipulation without sacrificing the inherent adaptability of the soft structure.

To further enhance the effectiveness and reliability of the perception strategy, we propose a deformable perception method capable of force and pose estimation using a single internal vision, as shown in Fig. 1. We advocate a systematic approach that enables in-finger manipulation using a soft active surface empowered by multi-modal vision-based deformable perception without relying on multiple sensors. The main contributions and structure of this paper are as follows:

- Design and fabrication of a compact ActiveSPN2 module with integrated inner vision feedback, with in-finger dexterity while preserving the adaptability of the soft polyhedral network.
- Propose a novel multi-modal perception architecture for ActiveSPN2 that leverages vision-based deformable perception to estimate both pose and force, enabling in-finger manipulation without relying on multiple sensors.
- Closed-loop in-finger manipulation demonstration: the ActiveSPN2 gripper is capable of real-time estimation and manipulation of object pose and force, enabling robust perception and dexterous operation within confined spaces.

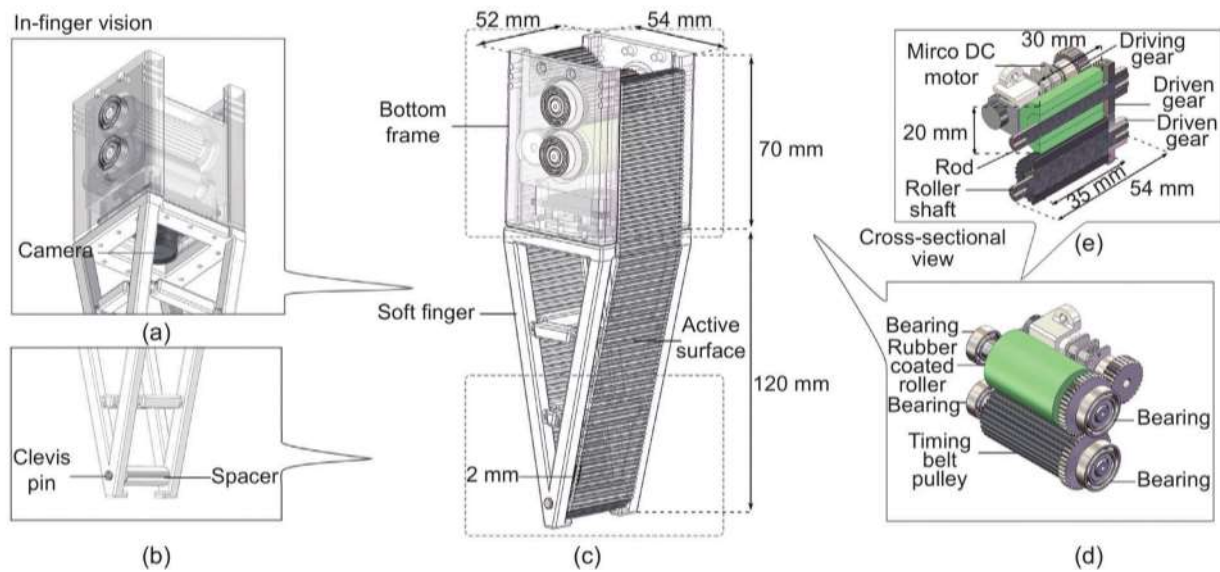


Fig. 2. Structural Design and Component Details of an ActiveSPN2 Module. (a) In-finger vision. (b) Fingertip, inside the soft active surface. (c) Whole ActiveSPN2. (d) Physical actuation system. (e) Cross-sectional view of the physical actuation system.

Details of the design and model of the ActiveSPN2 module and gripper are discussed in the following section. The architecture of the force and pose estimation is discussed in Section 3. The experiments implemented are discussed in Section 4. Finally, concluding remarks are made in Section 5.

2. ActiveSPN2 as soft active surfaces

To verify the hypothesis that it is possible to perform force and pose estimation on a soft active surface, a vision-based deformable approach was demonstrated by building the ActiveSPN2 prototype, as described in this section. Details of both the simplified one-finger model and the three-finger model are also discussed.

2.1. Specifications of ActiveSPN2 module

This study introduces a multi-modal vision-based deformable perception approach that uses a single in-finger vision system to infer the grasping state of the soft active surface, including force and pose estimation. Our module design, ActiveSPN2, is shown in Fig. 2. The ActiveSPN2 module comprises three main components: the in-finger vision part for internal visual perception, the body part for structural stability and 3D omnidirectional adaptability, and the physical actuation system for actuating the active surface.

ActiveSPN2 is a completely new design compared to our previous prototype, ActiveSPN [21,32], with a total weight of 280.5 g and dimensions of $54 \times 52 \times 190$ mm. The main improvements are as follows: (1) a markerless deformable perception method was implemented to enable real-time force perception; (2) a rubber-coated roller was added, forming a sandwich with the timing belt (active surface) pressed between the rubber-coated roller and the timing belt pulley, ensuring continuous and stable surface contact. This design enhances the gripper's ability to conform to varied object shapes without detachment or instability; (3) the timing belt pulley, the roller shaft, and other related components were redesigned and upgraded from 3D-printed parts to machined aluminum alloy parts, enabling more stable torque output; (4) additional bearings were incorporated at the contact interfaces between the rotating shafts of the rod and roller shaft and the support frame, in order to reduce friction.

About the in-finger vision part, as shown in Fig. 2(a), each finger module has an internal built-in camera unit, which is mounted on the bottom frame to capture real-time visual information from the soft polyhedral networks and information through the transparent timing belt. We employ an RGB camera, which offers a resolution of 640×360 at 330 Hz, with an FOV of 170° . This built-in camera enables continuous monitoring of the hand-held object's state and the segmented mask of the soft polyhedral networks throughout the manipulation process. Fig. 2(b) is an inside view below the soft active surface, which has a spacer to reduce friction when the soft active surface works.

The body part mainly consists of a bottom frame, a soft polyhedral network, and an active surface, as shown in Fig. 2(c). The bottom frame is fabricated from a photosensitive resin using stereolithography, with dimensions of $54 \text{ mm} \times 52 \text{ mm} \times 70 \text{ mm}$. It serves as a rigid foundation for supporting the soft polyhedral network and encloses the pulley mechanism used for belt-tension adjustment. The soft polyhedral network weighs approximately 40.2 g. Its length, width, and height are 54 mm, 52 mm, and 120 mm, respectively. It is fabricated from polyurethane elastomer (Hei-cast 8400 from H&K) through vacuum molding. The active surface is also fabricated using polyurethane elastomer, cast via vacuum molding. It features a hardness rating of 90A, a thickness of 2 mm, and is lightweight, weighing only 20.8 g. Its transparency allows for direct visual observation of the hand-held object during in-finger manipulation. The transparent timing belt is installed by wrapping it around the outer surface of the soft polyhedral network, with its tension finely adjusted via the timing belt pulley mechanism.

The physical actuation system is shown in Fig. 2(d), mainly including a Micro DC motor, a rubber-coated roller, a timing belt pulley, some gears, and bearings. The Micro DC motor is equipped with an encoder and communicates with the controller via a serial interface. The whole physical actuation system is compactly designed and weighs only 110.6 g. We adopted a high gear reduction ratio of 380:1, enabling a maximum effective torque of 3.4 kg cm, which allows the soft active surface to adjust the pose of the handheld object during grasping. The rubber-coated roller is 30 mm in length, 20 mm in diameter, and weighs 33.5 g. It is designed to form a sandwich with the timing belt pressed between the roller and the timing belt pulley, ensuring continuous and stable surface contact. The timing belt pulley was

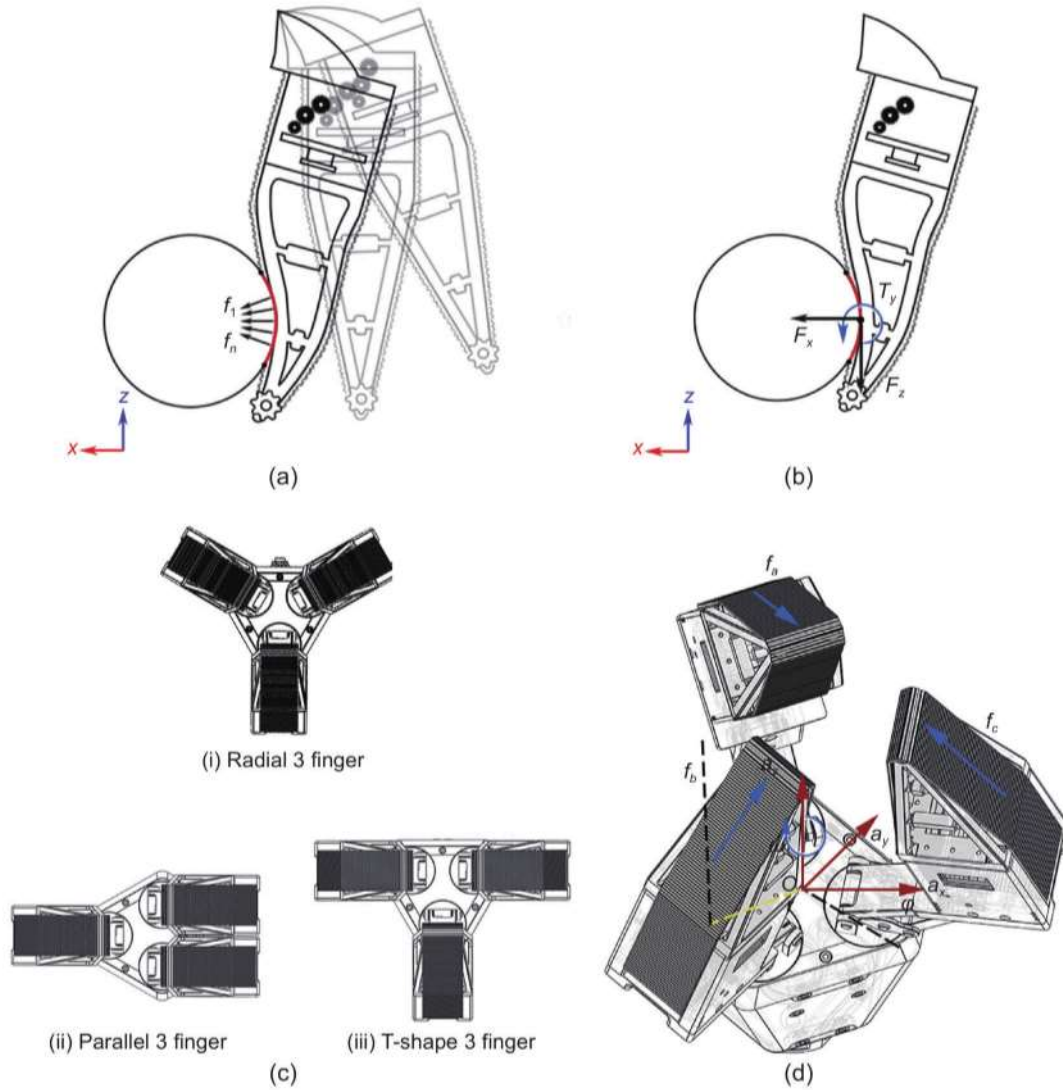


Fig. 3. Grasp modeling with ActiveSPN2. (a–b) Schematic of a single ActiveSPN2 module contact in X-Z cross-section. (c) Reconfigurable grasping of a 3-finger ActiveSPN2 gripper, including (i) Radial, (ii) Parallel, and (iii) T-shape Configuration. (d) Dynamic model of radial 3-finger configuration.

fabricated from 6061 aluminum alloy, treated with anodizing and sandblasting. It measures 35 mm in length and 18.52 mm in diameter, and weighs 15.4 g. The roller shaft, also made of 6061 aluminum alloy, measures 54 mm in length and weighs 8.8 g. An internal cross-sectional view of the physical actuation system is shown in Fig. 2(e), illustrating the internal structures of the roller shaft, timing belt pulley, and rubber-coated roller. These components are structurally matched to each other to enhance the stability of contact between the actively driven parts.

2.2. Dynamics of ActiveSPN2 module

In the simplified ActiveSPN2 module scenario illustrated in Fig. 3(a), the contact force is given by: $F = \sum_{j=1}^n f_j$. In the X-Z cross-section shown in Fig. 3(b), the model is further simplified. Then, the resultant contact force can be simplified to a normal force component F_n in the direction $\hat{x}_n$, $F_n = F_x > 0$. While the friction force, F_s , is the resultant of the components in the $\hat{y}_n$ direction and the $\hat{z}_n$ direction, given by, $F_s = F_{s,y} + F_{s,z}$, where $F_{s,y}$ and $F_{s,z}$ are the friction components in the $\hat{y}_n$ and $\hat{z}_n$ directions, respectively. Since the analysis is in the X-Z cross-section, we have $F_s = F_{s,z}$. Additionally, the contact generates a balancing torque component T_y along the $\hat{y}_n$ direction. When the object is

grasped, the shear force satisfies $\|F_s\| \leq \mu F_n$, $\dot{x} = 0$, where μ is the coefficient of friction and $\dot{x}$ is the sliding velocity of the object. If the sliding velocity is zero, the normal force is $F_n = F_x$, and the shear force is given by $F_s = F_{s,z}$.

When the gripper has three fingers, we can model the contact as three contact points. The dynamic equations of this system are:

$$\begin{aligned} \sum_i F_n^i + \sum_i F_s^i + F_{ext} - m_c g &= m_c \ddot{x}, \quad (i = 1, 2, 3) \\ \sum_i T_y^i + T_{ext} &= I \ddot{\theta}, \quad (i = 1, 2, 3) \end{aligned} \quad (1)$$

where F_n^i represents the normal force applied by the i th finger, F_s^i is the shear force applied by the i th finger, $F_s^i = F_{s,y}^i + F_{s,z}^i$, and F_{ext} is the external force. The mass of the grasped object is denoted as m_c , with its linear acceleration represented by $\ddot{x}$. T_y^i refers to the torque around the y -axis applied by the i th finger, while T_{ext} denotes the external torque. The moment of inertia of the hand-held object is I , and $\ddot{\theta}$ is the angular acceleration. Assuming the hand-held object is light such that its weight is neglected, the system satisfies the static equilibrium equations: $\sum_i F_n^i + \sum_i F_s^i + F_{ext} = 0$, $\sum_i T_y^i + T_{ext} = 0$, with the friction constraint $\|F_s^i\| \leq \mu F_n^i$, ($i = 1, 2, 3$).

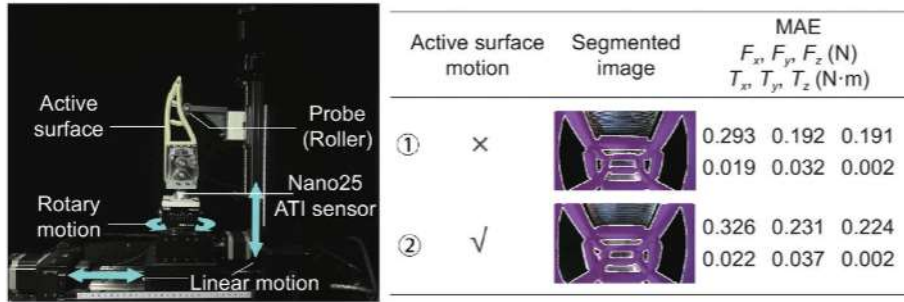


Fig. 4. Quantitative evaluation of force estimation with and without active surface motion: (Left) experimental setup. (Right) quantitative results.

2.3. Dynamics of ActiveSPN2 gripper

As shown in Fig. 3(c), the ActiveSPN2 can be configured in three ways: parallel 3-finger (P3), radial 3-finger (R3), and T-shaped 3-finger (T3). As illustrated in Fig. 3(d), a special case of the ActiveSPN2 is a radial 3-finger configuration, spaced 120° apart when designing the movement direction of each belt. Given that the mass of the hand-held object is m , with the three ActiveSPN2 modules having mass m and moment of inertia $J = 2mr^2$, the dynamic equations of the system can be expressed as follows:

$$\begin{bmatrix} \dot{\omega}_x \\ \dot{\omega}_y \\ a_z \end{bmatrix} = \begin{bmatrix} \frac{-1}{m} & \frac{1}{2m} & \frac{1}{2m} \\ 0 & -\frac{\sqrt{3}}{2m} & \frac{\sqrt{3}}{2m} \\ \frac{r}{J} & \frac{r}{J} & \frac{r}{J} \end{bmatrix} \begin{bmatrix} f_a \\ f_b \\ f_c \end{bmatrix}, \quad (2)$$

where $\dot{\omega}_x, \dot{\omega}_y$ is the angular acceleration along the $\hat{x}_n, \hat{y}_n$ directions; a_z denote the linear acceleration components of the hand-held object along the $\hat{z}_n$ directions; ψ is the angular configuration of the system; r represents the length or characteristic dimension related to the kinematic setup; a_a, a_b and a_c are the accelerations of belts A, B, and C, respectively, and the corresponding forces are given by $f_a = ma_a, f_b = ma_b$, and $f_c = ma_c$.

After obtaining the dynamics model, we can identify the extremal distribution of accelerations across all directions, which corresponds to estimating the optimal actuation direction for forces and accelerations. The conclusion is that the maximum translational acceleration of the grasped object is achieved when moving the object along the axial direction of the finger.

3. Multi-modal VBDormP architecture

To predict force and the pose of the handheld object, we developed a multi-modal vision-based deformable perception framework that fully exploits in-finger visual information through a latent representation.

3.1. In-finger deformation tracking

In our previous studies on ActiveSPN [21], our focus was primarily on estimating the pose of the hand-held object. We observed the hand-held object's features under different poses through the transparent belt to make predictions. Therefore, for pose estimation, the learning strategy can be performed directly on the raw inner images. In contrast, force perception centers on how the soft polyhedral networks deform under different object contact conditions. Thus, it is necessary to achieve real-time segmentation and tracking of the adaptive deformation of the soft polyhedral network. In this study, we adopted the XMem model [33] to realize this, achieving real-time tracking of the SPN

deformation masks at a frame rate of 30 fps, as shown in Fig. 5(a). This implementation can be formally expressed as follows:

$$\mathbf{m}_n = f(\mathbf{x}_n), n \in \{1, 2, 3\} \quad (3)$$

where $\mathbf{m}_n$ represents the segmented mask in real time of the finger, $\mathbf{x}_n$ represents the real-time image of the n -th finger, and f represents the model of using XMem.

Mask noise from belt motion and visual noise may impair deformation perception and force estimation. To analyze noise induced by active surface motion, we performed the experiment shown in Fig. 4 using a custom test rig with an ATI Nano25 sensor for ground-truth measurement. The only variable was whether the active surface was driven or stationary. For each experimental condition, 1000 samples were collected, and the force estimation performance was quantitatively evaluated using the Mean Absolute Error (MAE). According to the result, force and torque MAEs increase slightly from (0.293, 0.192, 0.191) N and (0.019, 0.032, 0.002) N m when active surface is stationary to (0.326, 0.231, 0.224) N and (0.022, 0.037, 0.002) N m when active surface is moving, showing that ActiveSPN remains robust to belt motion. We further evaluate robustness to visual noise by adding square block noise of size $n \times n$ pixels ($n \in 0, 10, 20, 40, 80, 160$) into the segmentation masks and reporting force-torque MAE on the same 1000-sample test set. As shown in Table 2, even large noise blocks up to 160×160 pixels increase force MAEs by 0.2–0.5 N and torque MAEs by 0.01–0.02 N m, while errors remain small for noise blocks below 40×40 pixels. These results confirm that ActiveSPN effectively preserves meaningful deformation cues under both active surface motion and severe visual noise.

Moreover, in the proposed system, perception, inference, and control are implemented in a fully asynchronous multi-threaded architecture. While XMem determines the segmented mask update rate at 30 Hz, the VBDormP framework and the PD control loop are executed in parallel without blocking the mask tracking thread. In practice, each inference of the VBDormP framework takes only 2.84 ms on average, and the control computation introduces negligible additional delay. Therefore, the effective closed-loop latency is dominated by the 33 ms frame interval of XMem rather than the sum of all computational modules. This design ensures that the system maintains stable and responsive real-time control even under fast contact dynamics.

3.2. Multi-modal generative architecture

Our framework takes in-finger images as input and follows a growing research trend in which the captured image representation is treated as the latent variables of a generative model when projecting complex soft robotic deformations into image space [34]. To address this, we employ the Variational Autoencoder (VAE) framework, which offers structured and informative latent representations for downstream tasks. Based on

Table 2
Quantitative evaluation of force estimation under varying visual noise.

	Noise size	Mask before noise	Mask after noise	Segmented image	MAE F_x, F_y, F_z (N) T_x, T_y, T_z (N m)
①	0×0				0.286 0.186 0.186 0.018 0.030 0.002
②	10×10				0.291 0.187 0.187 0.018 0.030 0.002
③	20×20				0.302 0.190 0.190 0.019 0.032 0.002
④	40×40				0.334 0.210 0.199 0.020 0.036 0.002
⑤	80×80				0.441 0.305 0.240 0.031 0.050 0.003
⑥	160×160				0.886 0.658 0.376 0.077 0.108 0.007

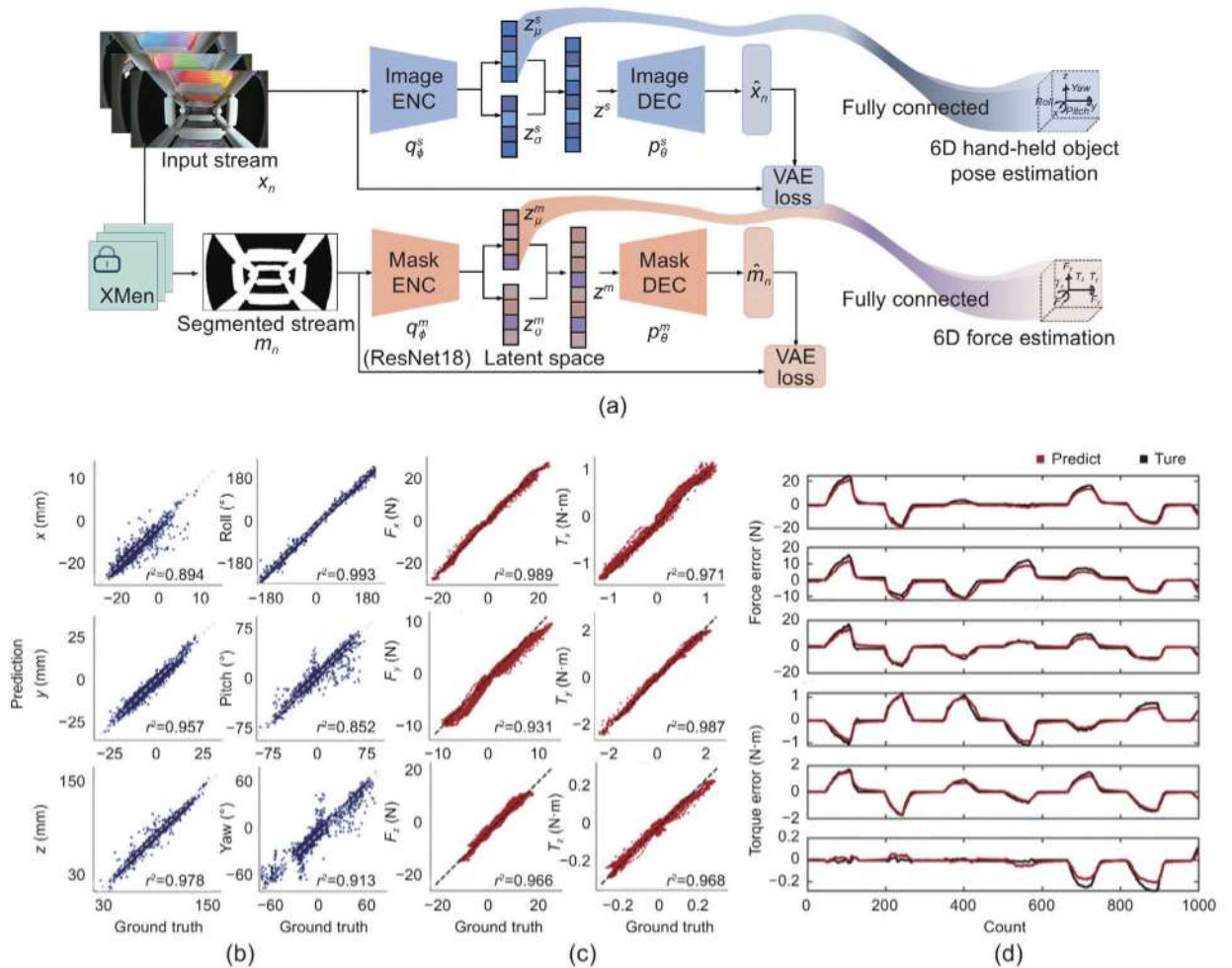


Fig. 5. Multi-Modal Vision-Based Deformable Perception Architecture and Prediction Performance. (a) Multi-Modal VBDormP architecture. Quantitative comparison between the predicted values and true labels obtained from (b) ArUco tags and (c) ATI nano25 force sensor. (d) Test results for 6D forces and torques in sequence number, with the red lines as predicted and black lines as true measurements.



Fig. 6. Experimental data collection setup. (a) Force sensing configuration. (b) Pose estimation configuration.

this generative learning architecture, we propose a multi-modal vision-based deformable perception framework that simultaneously estimates both force and pose, leveraging an encoder-decoder architecture [35] to extract latent representations, as illustrated in Fig. 5(a). The generative model, which includes force and pose encoders q_ϕ^m, q_ϕ^s to process images from in-finger vision, is used to generate the latent codes $\mathbf{z}^m, \mathbf{z}^s$, corresponding to the two modalities, respectively. Here, we added an auxiliary supervised regression task for 6D force and torque prediction based on $\mathbf{z}_\mu^m$, and 6D hand-held pose estimation based on $\mathbf{z}_\mu^s$. The encoder architecture is given by:

$$\begin{aligned} \mathbf{z}^m &= q_\phi^m(\mathbf{m}_n), \\ \mathbf{z}^s &= q_\phi^s(\mathbf{x}_n), n \in \{1, 2, 3\}. \end{aligned} \quad (4)$$

Note that, $q_\phi^k, k \in \{m, s\}$ is the probabilistic encoder resulting in a latent vector $\mathbf{z}^k, k \in \{m, s\}$, ϕ is the parameters of the encoder neural networks, which must be optimized during training. We let the variational approximate posterior be a Gaussian with a diagonal covariance:

$$\log q_\phi^k(\mathbf{z}^k | \mathbf{x}_k) = \log \mathcal{N}(\mathbf{z}^k; \mathbf{z}_\mu^k, \mathbf{z}_\sigma^k \mathbf{I}), k \in \{m, s\}. \quad (5)$$

Finally, through the generative decoder p_θ^m and p_θ^s , our model reproduces images of the interactions based on the learned model. The decoder architecture is given by:

$$\begin{aligned} \hat{\mathbf{m}}_n &= p_\theta^m, \\ \hat{\mathbf{x}}_n &= p_\theta^s(\mathbf{z}_n^s), n \in \{1, 2, 3\}, \end{aligned} \quad (6)$$

where p_θ^k is the decoder function, and $\hat{\mathbf{m}}_n$, and $\hat{\mathbf{x}}_n$ are the reconstructed output image of the finger, the decoder distributions $p_\theta^k(\mathbf{z}^k) \sim \mathcal{N}(\mathbf{z}^k; \mathbf{0}, \mathbf{I}), k \in \{m, s\}$ followed a normal distribution, θ represents the decoder network parameters, optimized during training.

The model's loss function for training consisted of three parts: reconstruction loss, latent representation regularization loss, and prediction loss.

$$\begin{aligned} \tilde{\mathcal{L}}(\theta, \phi; \mathbf{k}_n, \mathbf{Y}_n) &= \frac{\alpha}{1 + \alpha} \|\mathbf{k}_n - \hat{\mathbf{k}}_n\| + \frac{1}{1 + \alpha} \|\mathbf{Y}_n - \hat{\mathbf{Y}}_n\| \\ &\quad - \beta D_{KL}[\mathcal{N}(\mathbf{z}_\mu^k, \mathbf{z}_\sigma^k) \parallel \mathcal{N}(\mathbf{0}, \mathbf{I})], \end{aligned} \quad (7)$$

where α and β are two hyperparameters we introduced to adjust the relative importance between the image reconstruction

and regression prediction tasks during optimization. $\|\mathbf{k}_n - \hat{\mathbf{k}}_n\|$ represents the image reconstruction loss, $\|\mathbf{Y}_n - \hat{\mathbf{Y}}_n\|$ represents the prediction loss, and $D_{KL}[\mathcal{N}(\mathbf{z}_\mu^k, \mathbf{z}_\sigma^k) \parallel \mathcal{N}(\mathbf{0}, \mathbf{I})]$ is the latent representation regularization loss, which is used to avoid posterior collapse. By setting $\alpha \rightarrow \infty$ or $\alpha \rightarrow 0$, this formulation degenerates to the vanilla VAE and vanilla ConvNet baselines respectively, thereby providing a unified and fair comparison framework. We observed experimentally that when $\alpha \geq 1$, increasing α leads to smaller image reconstruction errors but a gradual decrease in prediction accuracy. Conversely, when $\alpha \leq 1$, the training becomes more focused on the prediction task, resulting in higher validation accuracy. Considering the trade-off between image reconstruction and prediction performance, we set $\alpha = 0.1$ and $\beta = 1$, which empirically provided a good balance between the two objectives.

3.3. Data collection platform

Fig. 6 illustrates the data-collection setup. For force sensing, data were collected separately for each finger of the ActiveSPN2 module. During contact between a finger and an external object, the segmented mask of the finger and the corresponding force measurements from the Nano 25 ATI sensor were recorded. For in-finger object pose estimation, video data from the three in-finger cameras were streamed to a laptop and continuously recorded throughout the motion. During data collection, a human operator reoriented the in-hand object using a joystick while simultaneously capturing the in-finger visual images and the corresponding object pose determined via the attached ArUco tag. We define the pose of frame O relative to frame S as $\mathbf{g}_{so}$. The transformations from the camera frame C to frames S and O are denoted by $\mathbf{g}_{cs}$ and $\mathbf{g}_{co}$, respectively. The desired pose is then computed as $\mathbf{g}_{so} = \mathbf{g}_{cs}^{-1} \cdot \mathbf{g}_{co} \in SE(3)$ [21].

Importantly, the force deformation dataset was collected without using any task objects. Instead, a thin probe is pushed against the soft finger to generate diverse contact conditions and finger deformations, as shown in Fig. 7. The training data contain only finger-probe interactions, and the mask-only representation enables object-free force inference without relying on object geometry, texture, or shape. In addition, the model input consists solely of the segmented finger bone mask, as illustrated in Fig. 7,

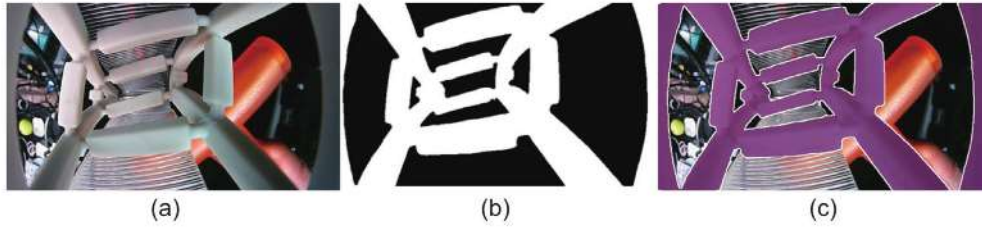


Fig. 7. Visualization of in-finger deformation tracking: (a) raw image captured by the in-finger camera, (b) the segmented finger's bone mask used as the actual input to the force estimation part, and (c) the mask overlaid on the raw image for deformation perception.

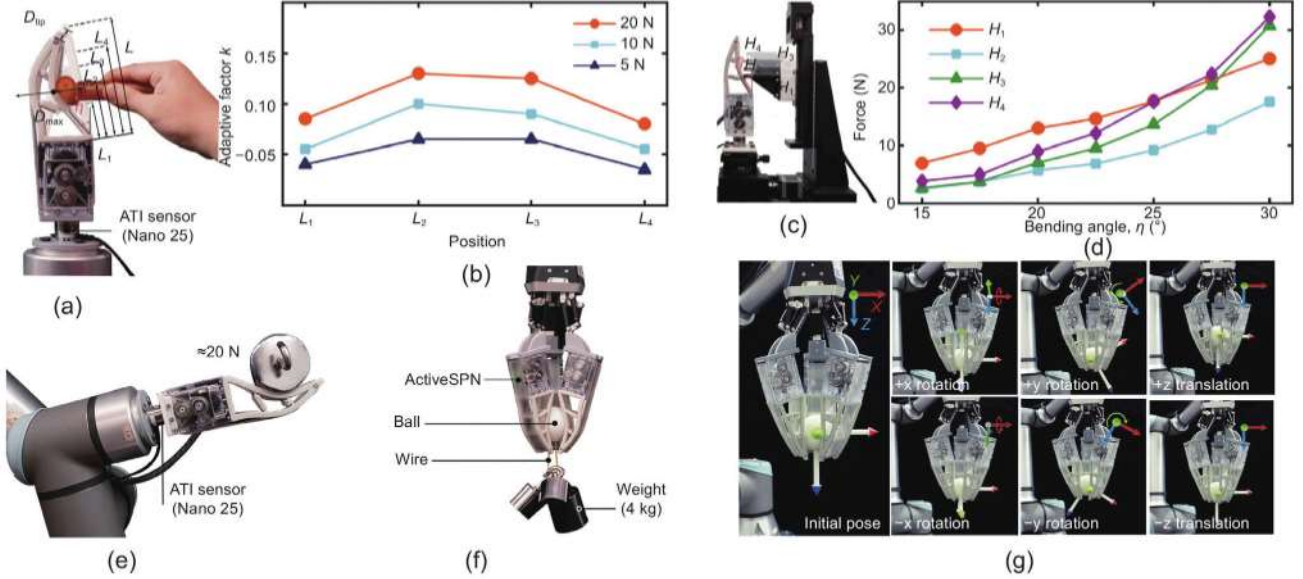


Fig. 8. Comprehensive Experimental Evaluation of Gripper Adaptability, Force Performance, and Manipulation Capability. (a–b) Measurement of the adaptive kinesthesia for the ActiveSPN2. (c–d) Measurement of contact forces at various bending angles and contact positions. (e) ActiveSPN2 weighing a 20 N mass. (f) Payload experiment setup demonstrating the gripper lifting a 4 kg weight. (g) Orientational object in-finger manipulation tests involving rotations and translations of the x-, y-, and z-axes.

which excludes all background and object appearance. As a result, the VBDormP framework learns the mapping from finger's geometric deformation to the resultant force, rather than overfitting to any particular object.

For training, the VBDormP framework was trained separately for force estimation and pose estimation tasks. For the force estimation network, training was conducted for 200 epochs with a batch size of 64 using the Adam optimizer and an initial learning rate of 0.001. The dataset was randomly divided into 8000 training samples and 2000 validation samples. To mitigate the potential inflation of validation accuracy due to sample dependence, an independent test dataset comprising 1000 samples was additionally collected. For the pose estimation network, each object category dataset was similarly divided into 4000 training samples and 1000 validation samples. The model was trained for 200 epochs with a batch size of 32 using the Adam optimizer and an initial learning rate of 0.001. All networks were trained on a computer equipped with an NVIDIA GeForce RTX 3080 Ti GPU. Considering the relatively small dataset size, the learning rate was gradually decreased throughout training to enhance model convergence and generalization performance.

Fig. 5(b) and (c) show scatter plots comparing the predicted pose and force with the ground truth from the testing dataset, showing a close match with excellent R^2 scores over 0.852 and 0.931. Fig. 5(d) presents a time-series comparison between the predictions and ground truth from the test dataset.

4. Experiments and results

4.1. System characterization and performance

In this subsection, a series of tests was conducted to evaluate the proposed ActiveSPN2 system's characterization and physical performance, including the effects of contact position and force on finger bending and compliance, gripper structural reliability and object handling, as well as in-finger manipulation capabilities.

4.1.1. Quantification of adaptive kinesthesia under external loading

To investigate the adaptive behavior of the soft polyhedral network under external loading, we measured its deformation and bending responses at different force magnitudes and contact locations, aiming to quantify mechanical compliance and spatial adaptivity. As shown in Fig. 8(a), each soft polyhedral network and timing belt is fabricated from thermoplastic polyurethane elastomers. This material provides the soft polyhedral network with resultant spatial compliance when subjected to external loading, a characteristic we call the adaptive kinesthesia [36] of the soft polyhedral network. Following the definition, an adaptive factor, κ , is introduced to quantify this behavior. Specifically, when an external force f is applied at location l along the primary interaction face, κ is defined as:

$$\kappa(f, l) = \frac{D_{\max}(l') - D_{\text{tip}}}{L}, \quad (8)$$

where D_{max} is the maximum displacement of the adaptive deformation, D_{tip} is the tip displacement, l' is the location of maximum displacement, and L is the finger length. In the experiments, external forces of 5 N, 10 N, and 20 N were applied at four positions L_1 , L_2 , L_3 , and L_4 along the finger, while the response displacements were measured using an ATI Nano25 sensor.

As illustrated in Fig. 8(b), the adaptive factor κ varies with both the loading position and the applied force. For a given load, the adaptive factor at the middle segments L_2 , L_3 is significantly higher than that at the proximal L_1 or distal L_4 ends, indicating stronger compliance and bending response when the finger is loaded at its midsection. In particular, under a 20 N load at L_2 , the adaptive factor reaches its maximum value of approximately 0.13, corresponding to a maximum bending angle. Across all tested forces, κ increases with loading magnitude while exhibiting a consistent spatial distribution pattern, with the midsection showing higher values than both the proximal and distal ends. Compared with the soft polyhedral network without a timing belt, the ActiveSPN2 exhibits slightly smaller adaptive factor values, although the overall trend remains the same. These results demonstrate that the soft polyhedral network exhibits pronounced spatial adaptivity under external loading. Midsection loading induces greater bending angles and displacements, thereby enhancing the mechanical compliance of the finger during object interaction. This property provides critical support for achieving stable and adaptable grasping and manipulation.

4.1.2. Bending angle and gripping force

We conducted a series of compression tests to investigate the relationship between the bending angle and different contact locations under varying normal forces. As illustrated in Fig. 8(c), the soft ActiveSPN2 module was mounted on a custom test rig equipped with a high-performance force sensor, ATI Nano25, and two motorized linear stages. The probe compressed the ActiveSPN2 module horizontally at a rate of 5 mm/s to a predefined depth. Experiments were performed at four distinct contact heights.

The results in Fig. 8(d) indicate that for all contact heights, greater compression produces higher contact forces, which correspond to larger bending angles. At $\eta = 15^\circ$, the forces remain relatively small, with H_1 around 6.88 N, H_2 about 2.41 N, H_3 close to 2.59 N, and H_4 near 3.78 N. When η increases to 30° , the forces grow significantly, with H_4 reaching the highest value at approximately 32.27 N, followed by H_3 at 30.76 N, H_1 at 25.02 N, and H_2 at 17.52 N.

These observations suggest that when subjected to external loading, contact positions respond differently in terms of mechanical behavior. At smaller angles, H_1 exhibits slightly higher force, likely because of the greater structural stiffness near the base. As compression increases and the angle surpasses 25° , the upper contacts H_3 and H_4 gain the advantage of a longer moment arm and a larger contact wrap, which results in substantially higher reaction forces. In other words, bending is primarily driven by the applied load, while the contact height determines how strongly the reaction force grows with the angle. In practical terms, lower contacts such as H_1 are sufficient for gentle gripping, whereas higher contacts such as H_3 and H_4 are better suited for achieving stronger gripping forces. The gripping force used in high-efficiency operations is typically around 8 N, ensuring stable grasping without slippage. The recommended grasping force range is approximately 20 N, and the bending angle is preferably kept below 30° to ensure safe and stable operation. Excessive force can increase frictional resistance but may also cause irreversible structural damage.

4.1.3. Gripper structural reliability and object handling performance

We conducted experiments to assess the structural reliability of the gripper under large bending angles and to evaluate its object handling performance, including load-bearing capacity, resistance to disturbances, and the practical limits of object size and weight. To ensure reliable operation under large bending angles, the internal driving section employs an aluminum alloy timing belt pulley and a rubber-coated roller to hold the active surface between them like a sandwich. On the external side, where the ActiveSPN2 module contacts the object, the grasped object presses the timing belt against the soft polyhedral network, as shown in Fig. 8(e). Although curling of the soft polyhedral network creates a concavity, the belt is sandwiched between the soft polyhedral network and the object, thereby filling the concavity. Consequently, even under large bending angles, the belt neither loosens nor detaches. To further validate this observation, an object with a weight of 20 N was placed on the ActiveSPN2 module to induce bending. The results confirmed that, despite the formation of a concavity during bending, the belt remained securely sandwiched between the soft polyhedral network and the object, preventing belt loosening or detachment.

A payload test was conducted to evaluate the gripper's capability for lifting heavy objects and its resistance to external disturbances. The ActiveSPN2 gripper firmly grasped a ball with standard weights suspended below and gradually lifted weights up to 4 kg, as shown in Fig. 8(f). The weights, connected via a wire, introduced slight oscillations during the lift, causing shifts in the load direction. Despite these fluctuations, the ActiveSPN2 gripper maintained a stable grasp, demonstrating its ability to handle heavy objects and resist perturbing forces. This enhanced load-bearing capability is attributed to the flexible soft polyhedral networks, which conform to and partially wrap around the object, thereby increasing the contact area and providing additional mechanical support beyond static friction. However, the maximum weight of objects that the ActiveSPN2 gripper can practically manipulate is lower than this value, because the ActiveSPN2 gripper relies on continuous motor torque to maintain the gripping force during manipulation, which fundamentally limits its operational performance. Assuming an object is positioned at the midpoint between the three ActiveSPN2 module, the maximum manipulable weight can be estimated as $G_{max} = \frac{3\mu_b\tau_m}{L_c}$, where μ_{b0} is the friction coefficient between the timing belt and the hand-held object, $\tau_m = 3.4 \text{ kg cm cm}$ is the maximum motor torque, and L_c is the distance between the motor output shaft and the ActiveSPN2 contact point. Taking $\mu_b = 0.45$, the maximum manipulable weight is approximately 0.765 kg. Furthermore, the maximum and minimum sizes of objects that can be handled depend on the opening range of the DH-Robotics DH3 gripper's fingers, to which the ActiveSPN2 module is mounted, with sizes ranging from approximately 15 to 95 mm.

4.1.4. In-finger manipulation and object reorientation

To assess the dexterous manipulation capabilities of the ActiveSPN2 gripper, object manipulation experiments were conducted. During the experiments, a spherical object with a diameter of 70 mm was manipulated in-hand using the ActiveSPN2 gripper, allowing rotation around the x - and y -axes and translation along the z -axis, as shown in Fig. 8(g). In the initial posture prior to manipulation, the coordinate system of the object was aligned with that of the gripper along the x -, y -, and z -axes. In an unobstructed environment, the rotation range of the object around the x - and y -axes spans from -180° to 180° , while translation along the z -axis is allowed up to approximately 100 mm. To achieve any rotation transformation R of the object, $R \in SO(3)$, at least two rotations around non-redundant axes are required. By employing an extrinsic X-Y-X or Y-X-Y rotation sequence, which follows two common extrinsic Euler angle conventions, arbitrary rotations in $SO(3)$ can be accomplished.

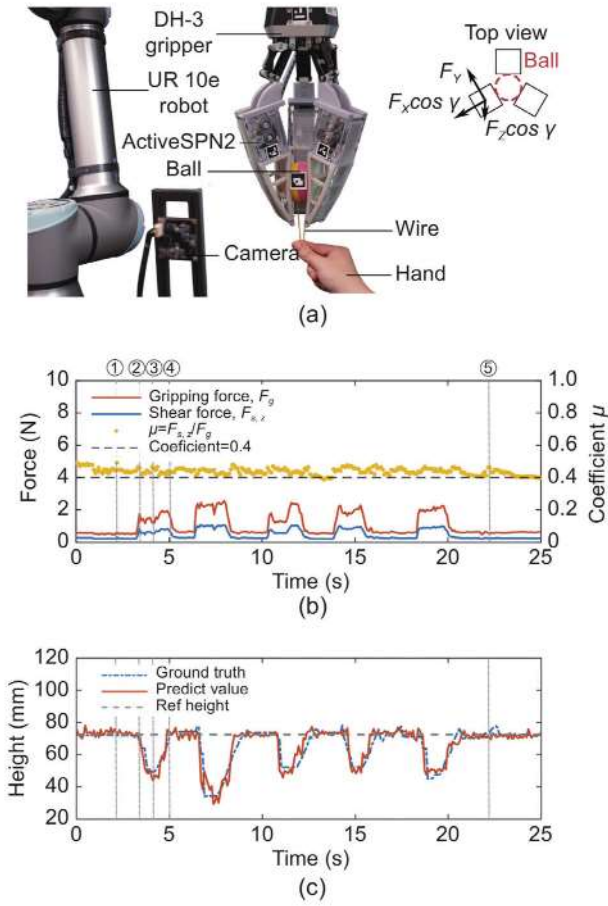


Fig. 9. Experimental Setup and Measurements of Human-Gripper Interaction. (a) Experimental setup of ActiveSPN2 interacting with a human hand. (b) Measured gripping, shear forces, and friction coefficient. (c) Time series of lift height. The key timing is as follows, initial state during ①, human hand contact and pulling down during ②–③, peak displacement at ③, active height recovery during ③–④, and stable grasp equilibrium at ⑤.

4.2. Validation of the multi-modal perception framework

To validate the effectiveness of the multi-modal perception framework, we conducted experiments including human-interactive grasping to assess the ActiveSPN2 gripper's perception capabilities, comparison with ATI ground-truth for force estimation accuracy, and peg-in tasks of light bulbs in confined spaces.

4.2.1. Competitive grasping with human interaction

To evaluate the superior multi-modal perception capabilities and the friction coefficient of the active surfaces in the proposed ActiveSPN2 in in-finger manipulation, we conducted competitive grasping experiments involving human interaction. As illustrated in Fig. 9(a), a DH-Robotics DH-3 three-finger gripper was employed, with its original rigid fingertips replaced by three ActiveSPN2 modules, and mounted on a UR10e collaborative robot. The experimental object was a spherical ball, which was connected to a human hand via a wire to perform competitive grasping tasks. During the experiment, the ActiveSPN2 gripper stably held the ball while the human hand applied a downward force by pulling the wire. When the ball was displaced from its initial height H_0 , the gripper, using pose estimation results and PD position control, actuated its deformable surface to actively reposition the ball to H_0 . Throughout the entire process, the gripping force $F_g = F_x \cos \gamma$, the shear force $F_{s,z} = F_z \cos \gamma$, the friction

coefficient $\mu = \frac{F_{s,z}}{F_g}$, and the ball's height were continuously recorded, where γ represents the inclination angle of the active surface with respect to the vertical plane.

The result of the gripping force and shear force is presented in Fig. 9(b), while the ball height is shown in Fig. 9(c). At time t_1 , the human hand had not yet contacted the wire, and the ball remained at its initial height H_0 , with both the gripping force and shear force stable at constant values. At time t_2 , when the human hand began the downward pull, both the gripping and shear forces increased simultaneously, while the ball height decreased. Under the action of these forces, the ball displacement reached its peak at time t_3 . Around this time, the human hand ceased pulling, and the ball reached its maximum height. After t_3 , the ActiveSPN2 gripper, under the PD controller, generated an upward motion by active surfaces to reposition the ball to H_0 . Under the action of the ActiveSPN gripper, the ball was gradually driven back to its initial position, with its height steadily recovering until stability was achieved at time t_4 . After five consecutive repetitions to verify system stability, the system reached quasi-static equilibrium at t_5 . The friction coefficient remained stable throughout the experiment, which is approximately 0.45.

4.2.2. Experimental validation of force estimation

To experimentally validate the perception accuracy of the proposed method in practical tasks, we selected the task of screwing in a light bulb. However, since the light bulb is monochromatic and lacks distinct features for pose estimation, this experiment primarily focuses on the results of force estimation. As shown in Fig. 10(a), in this experiment, a UR10e robotic arm equipped with an ActiveSPN2 gripper was used to screw the light bulb into its socket until the target torque was reached. An overhead camera tracked the yaw angle of the bulb using an ArUco tag and simultaneously recorded the estimated torque. An ATI Nano 25 force sensor mounted at the base of the bulb provided the ground-truth torque, while the predicted torque τ_c was computed from estimated force and torque measurements. Since the contact points $p_n, n \in \{1, 2, 3\}$ were not located at the center of mass c , the forces at these points need to be transformed into equivalent forces and torques at the center of mass according to $\tau_c = \sum_{n=1}^3 \tau_{s,z}^n + \sum_{n=1}^3 (p_n - c) \times F_{s,z}^n$, where $\tau_{s,z}$ are the estimated external torque from the ActiveSPN2 module, and $(p - c) \times F_{s,z}$ represents the equivalent torque at the center of mass.

The experimental results are shown in Fig. 10(b). At t_1 , the ActiveSPN2 gripper began to grasp the bulb, initiating contact, followed by the active surface's rotation during which the torque gradually increased. Once the maximum static friction threshold was exceeded, the bulb began to rotate. After t_2 , the bulb rotated, the yaw angle increased, and both the estimated torque and the ground-truth torque rose simultaneously. The predicted torque closely matched the ground truth throughout the screwing process, with both reaching the target value at approximately t_3 . After t_4 , the torques experienced oscillations due to overshoot from the PD torque controller. The curve indicated continuous bulb rotation until the target torque was achieved. At t_5 , the ActiveSPN2 gripper released the bulb. These results demonstrate that the proposed force estimation method achieves high accuracy in assembly tasks. The bulb screwing task was successfully completed in 7/10 trials, with the bulb rotating at an angular velocity of approximately 0.26 rad/s, and most failures occurring due to slight angular misalignments between the bulb and socket during initial contact. In the bulb-screwing experiments, the rotation speed ranged from approximately 0.20 to 0.39 rad/s. During this process, the measured torques gradually stabilized around 0.15 Nm. Although higher rotation speeds may induce additional inertial reaction forces, the velocities in our experiments were relatively low, and the torque differences between slow and fast rotations were negligible.

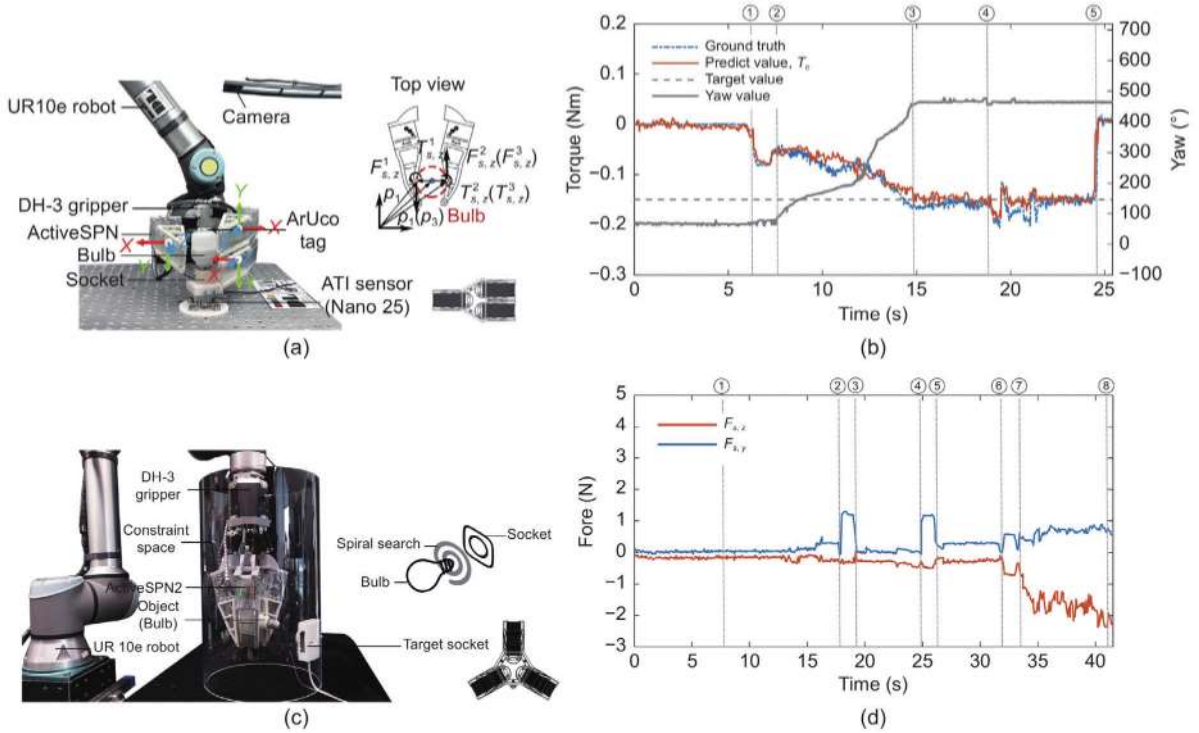


Fig. 10. Experimental Setup and Task Demonstrations of ActiveSPN2. (a) Experimental setup of ActiveSPN2 performing a light bulb screwing task. (b) Measured torque, ground-truth torque from the Nano25 sensor, and yaw angle estimated using the external camera by ArUco tag. (c-d) Setup and results of industrial task demonstrations verifying the applicability of the proposed gripper, including light bulb installation using vision-based deformable perception. The key timing of (b) is as follows, contact occurs at ①, torque buildup during ①-②, bulb rotation begins at ③, controlled screwing process during ③-④, and stable release at ⑤. The key timing of (d) is as follows, moving downward to a constrained space during ①, first contact with the socket at ②, retracting from the socket at ③, second contact with the socket at ④, subsequent retraction at ⑤, aligning with the hole at ⑥, beginning to tighten at ⑦, and final insertion of the bulb until it lights up at ⑧.

4.2.3. Peg-in task demonstrations in confined spaces

To validate the practical applicability of ActiveSPN2 in confined spaces, this section presents a light bulb peg-in task. As shown in Fig. 10(c), a UR10e robotic arm with an ActiveSPN2 gripper performed a spiral search [37], using the estimated shear force to identify the socket contact point while holding the light bulb. When the measured shear force was below a preset threshold, the position was considered a potential socket location, and the robotic arm moved further. Subsequently, under the control of ActiveSPN2, the bulb was screwed in.

The peg-in position was updated according to the spiral search. The goal of the spiral search is to locate the hole, as shown in Fig. 10(c). During the search, the ActiveSPN2 gripper holding an object moves along a specified vector, $\omega = \hat{y}_n \in \mathbb{R}^3$, until the following condition is no longer satisfied: $\hat{y}_n \cdot F_{s,y} \leq C$, where $F_{s,y}$ is the force obtained by the ActiveSPN2 gripper, C is a preset threshold chosen to be larger than the sensor noise.

Assume that the current position of the ActiveSPN2 gripper is q_b . During the spiral search, the ActiveSPN2 gripper moves along spiral curves on the X-O-Z plane, whose normal vector is ω . The position during the spiral search is updated according to the following equation:

$$p_{j+1} = r_{j+1} \cdot e^{\hat{\omega}\theta_{j+1}} \cdot (I_x + I_z) + p_j \quad (9)$$

where p_i is the current position, $r_{j+1} = r_j + \delta r$, $\theta_{j+1} = \theta_j + \delta\theta$, $\delta\theta$ and δr representing the spiral step sizes. The term $e^{\hat{\omega}\theta}$ corresponds to Rodrigues' formula, given by $e^{\hat{\omega}\theta} = I + \hat{\omega} \sin \theta + \hat{\omega}^2(1 - \cos \theta) \in SO(3)$, and I_x, I_z are unit vectors along the X and Z axes, respectively.

The experimental results are shown in Fig. 10(d). Initially, a 45-mm-diameter light bulb is grasped by the ActiveSPN2 gripper, which holds the bulb by its spherical part. At time t_1 , the ActiveSPN2 gripper held the bulb and moved downward slowly.

Until reaching the coarse positioning of the socket, it initiated the spiral search. At t_2 , after advancing toward the first search point, a large contact force was detected, indicating an obstacle; the gripper then retracted at t_3 . At t_4 , the second search point was tested and subsequently retracted. At t_5 , the third search point was tested, where the contact force was below the threshold; the arm then advanced a short distance. Finally, the bulb was screwed in until it lit up. This experiment demonstrates that the system's force-sensing capability has potential applicability in confined spaces. The insertion task was successfully completed in 6/10 trials, demonstrating the reliability of the spiral search and force-sensing control strategy. Most failures were due to lateral misalignment or insufficient friction compensation during the insertion process.

4.3. Demonstration of in-finger dexterity and robustness

Inspired by the manipulation task taxonomy, we also performed nonprehensile surface translation and in-finger manipulation of flexible objects.

Non-prehensile manipulation, as illustrated in Fig. 11(a), begins with the ActiveSPN2 gripper opened in T-shape configuration and the ActiveSPN2 applying a downward force on the pencil surface. Once the pencil is pressed against the table, moving the right finger belts toward the inside and the left finger belts away from the inside causes the pencil to slide to the left. This strategy is particularly suitable for manipulating slender objects that are difficult to grasp securely. Notably, the end effector remains stationary, and the object moves through frictional interactions with the belts and the supporting surface. This task was successfully completed in 9/10 trials.

As shown in Fig. 11(b), the cloth is held in-hand using the ActiveSPN2 gripper, with direct contact between the cloth and

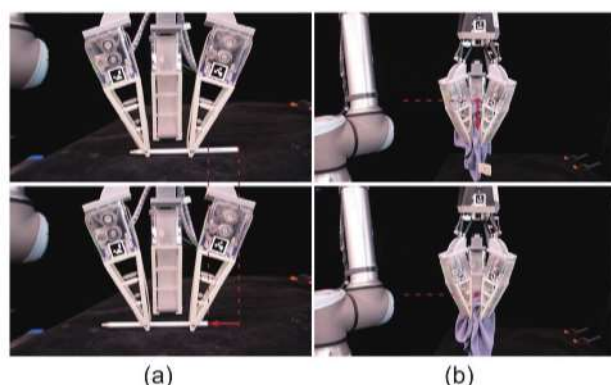


Fig. 11. The ActiveSPN2 design enables versatile manipulation beyond conventional grasping: (a) non-prehensile manipulation demonstrated by sliding a pencil across the table, and (b) grasping of deformable cloth to validate adaptability to flexible objects.

the ActiveSPN2 module. Excessively tight grasps hinder internal manipulation of the cloth, while overly loose grasps cause the soft material to slip easily. The high adaptivity of the ActiveSPN2 gripper mitigates these issues more effectively than rigid grippers. By executing a Z-translation motion, the cloth can be transported vertically within the hand, demonstrating controlled in-finger manipulation. The cloth manipulation task achieved a success rate of 6/10 trials, with failures mainly due to insufficient friction between the cloth and the gripper surface during vertical translation.

5. Conclusion

This study presents ActiveSPN2 as a novel gripper that achieves multi-modal deformable perception on a soft active surface without relying on multiple sensors. The feasibility of this multi-modal perception strategy, which provides 6D force and object pose estimation for the soft active surface, was validated through a series of manipulation tasks using the ActiveSPN2 gripper. Experimental results demonstrate that, during in-finger manipulation, high-precision real-time multi-modal deformable perception can be achieved.

The contributions of this study are as follows: we designed and fabricated a compact ActiveSPN2 module with integrated in-finger vision feedback, achieving dexterous in-finger manipulation while preserving the adaptability of the soft polyhedral network; we proposed a novel multi-modal perception architecture that leverages vision-based deformable sensing to estimate both pose and force, enabling in-finger manipulation without relying on multiple sensors; finally, we demonstrated closed-loop in-finger manipulation, showing that the ActiveSPN2 gripper can perform real-time estimation and control of object pose and force, enabling robust perception and dexterous operation in confined spaces. The proposed perception method enhances the soft active surface's perception capability and improves the success rate of in-finger manipulation in constrained environments.

Future work will focus on three aspects. Structurally, we aim to make the sensor and actuated fingers more compact. In terms of perception, we plan to incorporate additional perception modalities during manipulation to improve estimation capabilities. Algorithmically, we will integrate advanced learning-based estimation techniques and extend the approach to more complex multi-contact manipulation scenarios. These planned improvements are expected to lay the groundwork for applying our framework to large-scale, unstructured environments such as warehouse operations and industrial maintenance tasks, thereby bridging the gap between laboratory settings and practical deployment.

CRedit authorship contribution statement

Sen Li: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation. **Chaoyang Song:** Writing – review & editing, Resources, Methodology, Conceptualization. **Fang Wan:** Writing – review & editing, Visualization, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.birob.2026.100326>.

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